Exact Optimal Cloning of Stabilizer States, and the Price of Measuring First

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Abstract

We determine the optimal average global fidelity of $T \to T+k$ approximate cloning for the uniform ensemble of n-qubit stabilizer states, for every number of qubits n, every $k \geq 1$ and every $1 \leq T \leq 3$. With $D = 2^n$, the optimum is

$$P_{\mathrm{stab}}(n;T,k) = \binom{D+T-1}{T} F_{T+k-\kappa(k)}, \qquad F_s = \frac{\prod_{i=0}^{s-2}(2^i+1)}{D\prod_{i=0}^{s-2}(D+2^i)},$$

where F_s is the stabilizer frame potential and $\kappa(k)$ is the largest dimension of a doubly-even binary code of length k. The result rests on an exact formula for the largest eigenvalue of the mixed stabilizer moment operators: for all $a+b \geq 1$ and all n it equals F_{a+b-w}, where $w = \min(a,b) + \kappa(|a-b|)$ is the Witt index of an associated $\mathbb{Z}_4$-valued quadratic form. We then ask what is lost by *measuring first*. For measure-and-prepare (entanglement-breaking) cloners we show that, after the pretty-good measurement, re-preparing a better output than copies of the guess helps exactly from four output copies on, because fourth powers erase the phases $\pm 1, \pm i$ of stabilizer amplitudes; that for $1 \to 4$ the optimum is $3\big(D+48+\sqrt{D^2+16D+1664}\big)/\big((D+1)(D+2)(D+4)(D+8)\big)$ for every n, so coherent cloning wins by less than a factor 5; and that for $2 \to 4$ there are two regimes: the pretty-good-measurement strategy is optimal for $n \leq 3$, while for every $n \geq 4$ Bell sampling is optimal, with fidelity $3/(D(D+2))$ and a coherent advantage of exactly $5D/(D+4)$. The upper bounds are explicit dual certificates for the PPT relaxation, covering every n.

1 Introduction

Werner determined the optimal global fidelity of cloning an unknown pure state [1]. For stabilizer states, Herasymenko et al. observe, in a trace-distance-oriented discussion, that the optimal cloning channel remains an open question [2], and Bansal, Caro and Mahajan establish the optimal constant-error cloning sample complexity $\Theta(n)$ [3]. Separately, Jeon, Sohn and Oh identify exact optimal fidelities beyond asymptotic scaling as an open problem for general mixed-state cloning [4]. These results address different figures of merit or parameter regimes from the finite-copy average global fidelity considered here. Since stabilizer states form a complex projective 3-design but not a 4-design [5, 6], Werner's universal cloner is optimal for them exactly when $T+k \leq 3$ [7]. To our knowledge, no general closed formula had been established for the optimal average global fidelity of uniform n-qubit stabilizer-state cloning throughout the non-design regime $T+k \geq 4$, for arbitrary n and all $1 \leq T \leq 3$.

We determine them. For $1 \le T \le 3$, every k and every n, the optimal average global fidelity is the closed form in the abstract (Corollary 11); Table 1 lists examples. The formula has two ingredients: the frame potential F_s of random stabilizer states [8][5, Thm. 2], and the coding-theoretic quantity $\kappa(k)$, the largest dimension of a doubly-even binary code of length k. The appearance of κ has a precise source: it is the Witt index of the $\mathbb{Z}_4$-valued quadratic form that indexes the relevant stabilizer moment.

Table 1: Optimal $T \to T + k$ cloning fidelity of uniform n-qubit stabilizer states (Corollary 11), with Werner's universal value in parentheses.

	$n = 1$	$n = 2$	$n = 3$	$n = 4$	$n = 5$
$1 \to 4$	5/12 (2/5)	1/8 (4/35)	1/36 (4/165)	1/204 (4/969)	5/6732 (4/6545)
$1 \to 5$	5/12 (1/3)	1/8 (1/14)	1/36 (1/99)	1/204 (1/969)	5/6732 (1/11781)
$1 \to 6$	3/8 (2/7)	3/32 (1/21)	1/64 (2/429)	1/544 (2/6783)	1/5984 (2/145299)
$2 \to 4$	5/8 (3/5)	5/16 (2/7)	1/8 (6/55)	1/24 (2/57)	5/408 (6/595)
$2 \to 5$	9/16 (1/2)	15/64 (5/28)	9/128 (1/22)	1/64 (1/114)	3/1088 (1/714)
$3 \to 4$	5/6 (4/5)	5/8 (4/7)	5/12 (4/11)	1/4 (4/19)	5/36 (4/35)
$3 \to 5$	3/4 (2/3)	15/32 (5/14)	15/64 (5/33)	3/32 (1/19)	1/32 (1/63)

Measuring first. Part II asks how much of this requires coherent processing. A measure-and-prepare cloner can measure the inputs, guess the state, and prepare copies of the guess, achieving $D_T F_{2T+k}$; asymptotic equivalence of cloning and estimation is a classical theme [9, 10], and re-preparing entangled outputs instead of identical copies is known to help asymptotically [10]. For stabilizer states we obtain the exact finite-copy picture (Table 2): for the pretty-good measurement, the gain from better re-preparation starts at exactly four output copies (Theorem 13); for $1 \to 4$ the optimal measure-and-prepare fidelity has a closed form for every n (Theorem 15); and for $2 \to 4$ the optimal measurement changes from the pretty-good measurement to Bell sampling [11] between three and four qubits (Theorem 18). In both cases coherent cloning beats any measure-and-prepare cloner by less than a factor of 5.

Table 2: Optimal coherent cloning P_{stab}, optimal measure-and-prepare cloning P_{EB}, and copying the guess of the pretty-good measurement P_{PGM}, for $1 \to 4$ and $2 \to 4$. For $2 \to 4$, P_{EB} is attained by the pretty-good-measurement strategy for $n \le 3$ and by Bell sampling for $n \ge 4$.

		$n = 1$	$n = 2$	$n = 3$	$n = 4$	$n = 5$
$1 \to 4$	P_{stab}	5/12	1/8	1/36	1/204	5/6732
	P_{EB}	$(5+\sqrt{17})/24$	$(13+\sqrt{109})/240$	$(7+\sqrt{29})/720$	$(8+\sqrt{34})/6120$	$(2+\sqrt{2})/13464$
	P_{PGM}	3/8	3/32	1/64	1/544	1/5984
$2 \to 4$	P_{stab}	5/8	5/16	1/8	1/24	5/408
	P_{EB}	$(9+\sqrt{65})/32$	$(41+\sqrt{1321})/384$	$(7+\sqrt{37})/256$	1/96	3/1088
	P_{PGM}	17/32	51/256	51/1024	17/2048	1/1024

Method. For $T \le 3$, Clifford symmetry reduces cloning to the largest eigenvalue of a mixed moment operator (Section 2). Gross, Nezami and Walter express stabilizer moments as sums over stochastic Lagrangian relations [8]; a partial transpose gives the mixed moments (Section 3). We compute the top eigenvalue for every signature by passing to the abstract algebra of relations, where

left multiplication by the moment is a nonnegative matrix whose column sums depend only on a "defect level" and are given exactly by frame potentials (Section 5). The top level is the Witt index, computed in Section 4. Related exact work by Hayashi and Lu diagonalizes the stabilizer Gram operator for state identification [12]; our argument does not use their spectral theorem. The moment identities [8][12, Lemma 6] and the frame potentials are known; what is new here is the top eigenvalue of the mixed moments (Theorem 10) and its consequence for cloning (Corollary 11).

Part I. Coherent cloning

2 Setting and reduction

Let $D = 2^n$, let $\mathrm{Stab}(n)$ be the set of n-qubit stabilizer states, $N_n = |\mathrm{Stab}(n)| = 2^n \prod_{i=1}^{n}(2^i+1)$, and $D_m = \binom{D+m-1}{m} = \dim \mathrm{Sym}^m(\mathbb{C}^D)$. A $T \to T+k$ cloner is a channel Φ from operators on $\mathrm{Sym}^T(\mathbb{C}^D)$ to operators on $\mathrm{Sym}^{T+k}(\mathbb{C}^D)$. This loses no generality: all inputs lie in $\mathrm{Sym}^T(\mathbb{C}^D)$, and projecting the output of an unrestricted channel onto $\mathrm{Sym}^{T+k}(\mathbb{C}^D)$, replacing the complementary outcome by a fixed state in that subspace, gives a channel whose fidelity with every $|S\rangle^{\otimes(T+k)}$ is no smaller. Then

$$P_{\mathrm{stab}}(n;T,k) = \max_{\Phi}\ \mathbb{E}_S \langle S^{\otimes(T+k)} | \Phi\big(|S\rangle\langle S|^{\otimes T}\big) | S^{\otimes(T+k)} \rangle,$$

with S uniform on $\mathrm{Stab}(n)$. For $a, b \geq 0$ define the mixed moment

$$X^{(a,b)} = \mathbb{E}_S\, |S\rangle\langle S|^{\otimes a} \otimes |\bar{S}\rangle\langle \bar{S}|^{\otimes b}.$$

Lemma 1 (Reduction). *For $1 \leq T \leq 3$ and $k \geq 1$, $P_{\mathrm{stab}}(n;T,k) = D_T\, \lambda_{\max}\big(X^{(T+k,T)}\big)$. Moreover the optimum is attained by a Clifford-covariant channel, which has the same fidelity on every stabilizer state, so the average and worst-case optima coincide.*

Proof. Fix an orthonormal basis $\{e_c\}$ of $\mathrm{Sym}^T(\mathbb{C}^D)$ and write $s_c(S) = \langle e_c | S^{\otimes T}\rangle$. With the Choi operator $J = \sum_{c,c'} \Phi(|e_c\rangle\langle e_{c'}|) \otimes |c\rangle\langle c'|$, the average fidelity is $\mathrm{tr}(JX)$ with $X = \mathbb{E}_S vv^\dagger$, $v = S^{\otimes(T+k)} \otimes \sum_c \overline{s_c(S)}|c\rangle$. This X is the compression of $X^{(T+k,T)}$ to $\mathrm{Sym}^{T+k} \otimes \overline{\mathrm{Sym}^T}$, which contains its support, so the two have the same nonzero spectrum. By semidefinite-programming duality (a strictly feasible point exists), $P_{\mathrm{stab}} = \min\{\mathrm{tr}\, Y : I \otimes Y \succeq X\}$, the minimum being attained. For a Clifford unitary C, X is invariant under $C^{\otimes(T+k)} \otimes \bar{C}^{\otimes T}$, so averaging an optimal Y over the Clifford group gives an optimal Y commuting with the action of $\bar{C}$ on Sym^T. For $T \leq 3$, the unitary 3-design property of the multiqubit Clifford group [13, 14] implies that its twirl on $\mathrm{End}((\mathbb{C}^D)^{\otimes T})$ equals the Haar twirl; an operator commuting with the Clifford action on $\mathrm{Sym}^T(\mathbb{C}^D)$, extended by zero, is therefore fixed by the Haar twirl, and irreducibility of $\mathrm{Sym}^T(\mathbb{C}^D)$ under $\mathrm{U}(D)$ forces it to be scalar. Thus $Y = \lambda I$. The constraint is then $\lambda \geq \lambda_{\max}(X)$, giving $P_{\mathrm{stab}} = D_T \lambda_{\max}(X)$. Averaging an optimal channel over the (transitive) Clifford action gives a covariant optimal channel with equal fidelity on every stabilizer state. □

3 Mixed moments as sums over Lagrangian relations

Write $t = a + b \geq 1$ and $V = \mathbb{F}_2^t$ with coordinates split into A (the first a) and B (the last b). Define the $\mathbb{Z}_4$-valued form $q_{a,b}(x) = |x_A| - |x_B| \bmod 4$, where $|\cdot|$ is the Hamming weight, and the dot product $\beta(x,y) = x \cdot y \bmod 2$. Then $q(x+y) = q(x) + q(y) + 2\beta(x,y) \pmod 4$.

Definition 2. $\Sigma(q_{a,b})$ is the set of t-dimensional subspaces $U \subseteq V \oplus V$ such that $q_{a,b}(x) \equiv q_{a,b}(y) \pmod 4$ for all $(x,y) \in U$, and $\mathbf{1}_{2t} = (\mathbf{1}_t, \mathbf{1}_t) \in U$. For $U \in \Sigma$ let $r(U) = \sum_{(x,y)\in U} |x\rangle\langle y|$ on $(\mathbb{C}^2)^{\otimes t}$, $R(U) = r(U)^{\otimes n}$ (acting on $(\mathbb{C}^D)^{\otimes t}$ after reordering), $N_L(U) = \{x : (x,0) \in U\}$ and $N_R(U) = \{y : (0,y) \in U\}$.

Every $U \in \Sigma$ is totally isotropic for $q \oplus -q$, hence for the polar form $\beta \oplus \beta$, and has dimension t, so $U = U^\perp$; in particular $\dim N_L(U) = \dim N_R(U)$, and both are totally isotropic for $q_{a,b}$. For $b = 0$, $\Sigma(q_{t,0})$ is the set $\Sigma_{t,t}$ of Gross–Nezami–Walter [8, Def. 4.1].

Lemma 3 (Mixed moments)**.** *For $t = a + b \geq 1$ and $n \geq 1$, $X^{(a,b)} = Z_{n,t}^{-1} \sum_{U \in \Sigma(q_{a,b})} R(U)$ with $Z_{n,t} = D \prod_{i=0}^{t-2} (D + 2^i)$.*

Proof. For $b = 0$ this is [8, Thm. 5.3], valid for all $n, t \geq 1$. In general $|\bar{S}\rangle\langle\bar{S}|$ is the transpose of $|S\rangle\langle S|$, so $X^{(a,b)}$ is the partial transpose of $X^{(t,0)}$ on the last b factors. On a single relation, $r(U)^{\Gamma_B} = \sum_{(x,y)\in U} |x_A y_B\rangle\langle y_A x_B| = r(U')$ with $U' = \{((x_A, y_B), (y_A, x_B))\}$. The map $U \mapsto U'$ preserves dimension and $\mathbf{1}_{2t}$, and $q_{a,b}(x_A, y_B) - q_{a,b}(y_A, x_B) = |x| - |y|$, so it is a bijection $\Sigma_{t,t} \to \Sigma(q_{a,b})$. The same statement appears as [12, Lemma 6]. □

Composition. For $U_1, U_2 \in \Sigma$ let $U_1 \circ U_2 = \{(x,z) : \exists y,\ (x,y) \in U_1, (y,z) \in U_2\}$ and $d(U_1, U_2) = \dim\big(N_R(U_1) \cap N_L(U_2)\big)$.

Lemma 4. *$r(U_1) r(U_2) = 2^{d(U_1,U_2)}\, r(U_1 \circ U_2)$ and $U_1 \circ U_2 \in \Sigma$. Hence $R(U_1)R(U_2) = D^d R(U_1 \circ U_2)$.*

Proof. The coefficient of $|x\rangle\langle z|$ in $r(U_1)r(U_2)$ counts y with $(x,y) \in U_1$, $(y,z) \in U_2$; this set is empty or a coset of $N_R(U_1) \cap N_L(U_2)$. The fibre product $\{(x,y,z)\}$ has dimension $2t - \dim(\pi_2 U_1 + \pi_1 U_2) = 2t - (t-d) = t + d$, using $(\pi_2 U_1)^\perp = N_R(U_1)$ and $(\pi_1 U_2)^\perp = N_L(U_2)$ (from $U = U^\perp$); its projection to $U_1 \circ U_2$ has kernel of dimension d, so $\dim U_1 \circ U_2 = t$. If y witnesses (x,z) then $q(x) - q(z) = (q(x) - q(y)) + (q(y) - q(z)) = 0$, and $\mathbf{1}_{2t} \in U_1 \circ U_2$ because it lies in both. □

Let $\mathcal{A}$ be the algebra with basis $\{e_U\}_{U\in\Sigma}$ and product $e_U e_{U'} = D^{d(U,U')} e_{U \circ U'}$. It has unit e_Δ for the diagonal $\Delta = \{(x,x)\} \in \Sigma$, and by Lemma 4 and bilinearity the map $\varphi(e_U) = R(U)$ satisfies $\varphi(uv) = \varphi(u)\varphi(v)$ and $\varphi(e_\Delta) = I$ (associativity of $\mathcal{A}$ is never used). Put $x = Z^{-1} \sum_U e_U$, so $\varphi(x) = X^{(a,b)}$, and let M be the matrix of left multiplication by x:

$$M_{U'',U'} = Z^{-1} \sum_{U:\ U\circ U' = U''} D^{d(U,U')} \ \geq 0.$$

Lemma 5 (Spectrum)**.** *$\lambda_{\max}(X^{(a,b)}) = \rho(M)$, the spectral radius of M.*

Proof. Since $M^j e_\Delta = x(x(\cdots(x e_\Delta)))$ and φ is multiplicative, $\varphi(M^j e_\Delta) = X^j$; so for the minimal polynomial p of M we get $0 = \varphi(p(M) e_\Delta) = p(X)$; hence every eigenvalue of X is a root of p, i.e. an eigenvalue of M, and $\lambda_{\max}(X) \leq \rho(M)$. Conversely, let $y \geq 0$, $y \neq 0$, satisfy $My = \rho(M) y$ (Perron–Frobenius for nonnegative matrices; no irreducibility is needed). Then $X\varphi(y) = \varphi(xy) = \rho(M)\varphi(y)$. Every $U \in \Sigma$ contains $(\mathbf{1}_t, \mathbf{1}_t)$, so $\langle \mathbf{1}_t^{\otimes n} | R(U) | \mathbf{1}_t^{\otimes n} \rangle = 1$ and $\langle \mathbf{1}_t^{\otimes n} | \varphi(y) | \mathbf{1}_t^{\otimes n} \rangle = \sum_U y_U > 0$. Hence $\varphi(y) \neq 0$ and $\rho(M)$ is an eigenvalue of X. □

4 The Witt index of $q_{a,b}$

Let $w(q)$ be the largest dimension of a subspace $W \subseteq V$ with $q|_W = 0$. Doubly-even binary codes of length m are exactly the totally isotropic subspaces of $q_{m,0}$, so $\kappa(m) = w(q_{m,0})$.

Proposition 6. *Let* $\mu(j) = 0,1,2,3,2,3,2,1$ *for* $j \equiv 0,\dots,7 \pmod 8$. *Then* $w(q_{a,b}) = \big(t - \mu(a-b)\big)/2$. *Equivalently* $\kappa(m) = 4\lfloor m/8 \rfloor + (0,0,0,0,1,1,2,3)_{m \bmod 8}$ *and* $w(q_{a,b}) = \min(a,b) + \kappa(|a-b|)$.

Proof. Upper bound. (i) $0 = q(2x) = 2q(x) + 2\beta(x,x)$, so $q(x) \equiv \beta(x,x) \pmod 2$. (ii) The Gauss sum factorizes over coordinates: $\gamma(q) = \sum_x i^{q(x)} = (1+i)^a(1-i)^b = 2^{t/2}\zeta_8^{a-b}$. (iii) If $q|_W = 0$ then $\beta|_W = 0$; for $x \notin W^\perp$, $\sum_{w\in W} i^{q(x+w)} = i^{q(x)} \sum_w (-1)^{\beta(x,w)} = 0$, while q is constant on cosets of W in $W^\perp$. Hence $\gamma(q) = |W|\gamma(\bar q)$ for the induced form on $W^\perp/W$, which has dimension $m = t - 2\dim W$, a nondegenerate polar form, and $\gamma(\bar q) = 2^{m/2}\zeta_8^{a-b}$. (iv) If $m = 0$ then $\gamma(\bar q) = 1$, forcing $a - b \equiv 0 \pmod 8$; if $m = 1$ the single basis vector has odd $\bar q$ by (i), so $\gamma(\bar q) = 1 \pm i$, forcing $a - b \equiv \pm 1$. With $m \equiv a - b \pmod 2$ this gives $m \ge \mu(a-b)$, i.e. $\dim W \le (t - \mu(a-b))/2$.

Lower bound. The words 1111; 11110; $\{111100, 110011\}$; the $[7,3]$ simplex code; and the extended Hamming $[8,4,4]$ code are doubly even, of dimensions $1,1,2,3,4$ at lengths $4,5,6,7,8$. Direct sums with the $[8,4,4]$ code give $\kappa(m+8) \ge \kappa(m) + 4$, so $\kappa(m) \ge (m - \mu(m))/2$, with equality by the upper bound. For $q_{a,b}$ with $a \ge b$, the b vectors $e_{A_i} + e_{B_i}$ ($q = 1 - 1 = 0$, disjoint supports) together with a maximal doubly-even code on the remaining $a - b$ coordinates of A span a totally isotropic subspace of dimension $b + \kappa(a-b)$; the case $a < b$ follows by exchanging A and B and replacing q by $-q$. Since $\kappa(m) = (m - \mu(m))/2$, this matches the upper bound. □

Remark. The appearance of doubly-even codes is not an accident of the proof. Polynomial invariants of the real and complex Clifford groups are spanned by complete weight enumerators of self-dual codes, doubly-even ones in the complex case [15]; complex-conjugate invariants, which are the ones relevant to $X^{(a,b)}$, are treated in [16] and, under stated hypotheses, through Γ-conjugate weight enumerators [17]. Proposition 6 gives this classical link an operational meaning: κ counts the copies that cloning gets for free.

5 The top eigenvalue

For $s \ge 1$ set $F_s = \prod_{i=0}^{s-2}(2^i+1)/\big(D\prod_{i=0}^{s-2}(D+2^i)\big)$, and write $(D)_k = \prod_{i<k}(D - 2^i)$, $c_k = \prod_{i=0}^{t-k-2}(2^i+1)$, and $\binom{r}{k}_2$ for the Gaussian binomial. The *level* of $U' \in \Sigma$ is $\dim N_L(U')$.

Lemma 7 (Column sums)**.** *The column of* M *indexed by* U' *sums to* $Z^{-1}\sum_{U\in\Sigma} D^{\dim(N_R(U)\cap N_L(U'))}$, *and if* U' *has level* r *this equals* $Z^{-1}S_t(r)$ *with* $S_t(r) = \sum_{k=0}^{r}\binom{r}{k}_2 (D)_k\, c_k$.

Proof. Each U contributes to exactly one row ($U'' = U \circ U'$), which gives the first expression. For a subspace W, $D^{\dim W} = |W|^n = \sum_{K\le W}(D)_{\dim K}$, by grouping n-tuples of vectors of W according to their span ($(D)_k$ counts surjections $\mathbb{F}_2^n \to K$, and vanishes when $k > n$). Exchanging sums, $\sum_U D^{\dim(N_R(U)\cap K')} = \sum_{K\le K'}(D)_{\dim K}\,\#\{U : K \le N_R(U)\}$ with $K' = N_L(U')$, and Lemma 8 evaluates the count as $c_{\dim K}$. □

Lemma 8 (Counting)**.** *For every subspace* $K \le V$ *of dimension* k *that is totally isotropic for* $q_{a,b}$, $\#\{U \in \Sigma : (0 \oplus K) \subseteq U\} = c_k$.

Proof. Let $J = \langle \mathbf{1}_{2t}\rangle \oplus (0 \oplus K)$. Elements of K have $q = 0$, hence even weight, so $(\beta\oplus\beta)(\mathbf{1}_{2t}, (0,h)) = 0$ for $h \in K$ and J is totally isotropic for $q \oplus -q$, of dimension $k+1$. On $J^\perp \subseteq \mathbf{1}_{2t}^\perp$ the form $q \oplus -q$ takes even values, so $Q = (q \oplus -q)/2 \bmod 2$ is an $\mathbb{F}_2$-quadratic form there with polar form $\beta \oplus \beta$; its radical on $J^\perp$ is J, so Q induces a nondegenerate quadratic form on $J^\perp/J$, of dimension $2m$ with $m = t - 1 - k$. Elements of Σ containing J correspond bijectively to Lagrangians (maximal totally singular subspaces) of $J^\perp/J$. A Lagrangian exists: $U_K = \{(u,v) : u, v \in K^\perp,\ u + v \in K\}$ has dimension t, contains J (since $\mathbf{1}_t \in K^\perp$), and satisfies $q(u) - q(u+h) = -q(h) - 2\beta(u,h) = 0$

for $h \in K$. A nondegenerate $2m$-dimensional $\mathbb{F}_2$-quadratic space containing a totally singular m-space is an orthogonal sum of m hyperbolic planes (choose e in it and f with $\beta(e,f)=1$, replace f by $f+Q(f)e$, and induct). In such a space the Lagrangians number $N(m)=\prod_{i=0}^{m-1}(2^i+1)$: with a hyperbolic pair (e_0,f_0) split off, one Lagrangian contains e_0 for each Lagrangian L' of the residual space, and those not containing e_0 are $\{x+\phi(x)e_0 : x\in L'\}\oplus\langle f_0+w+Q(w)e_0\rangle$ for a linear functional ϕ on L' and any w in the residual space with $\beta(w,x)=\phi(x)$ for all $x\in L'$ (such w exist by nondegeneracy, and the resulting Lagrangian does not depend on the choice). Thus $N(m)=(1+2^{m-1})N(m-1)$, $N(0)=1$, and the count is $N(t-1-k)=c_k$. □

Lemma 9 (A Gaussian-binomial identity)**.** $S_t(r)=c_r\prod_{i=t-r-1}^{t-2}(D+2^i)$. *Consequently* $Z^{-1}S_t(r)=F_{t-r}$.

Proof. Put $m=t-r-1$ and $a_j=\prod_{i<j}(2^{m+i}+1)$, so $c_k=c_r\,a_{r-k}$. It suffices to show $L_r:=\sum_k \binom{r}{k}_2(D)_k a_{r-k}=\prod_{i<r}(D+2^{m+i})$. Using $\binom{r}{k}_2=2^k\binom{r-1}{k}_2+\binom{r-1}{k-1}_2$, $a_{r-k}=a_{r-1-k}(2^{m+r-1-k}+1)$ and $(D)_k=(D)_{k-1}(D-2^{k-1})$, the two contributions to the coefficient of $\binom{r-1}{j}_2(D)_j a_{r-1-j}$ sum to $(2^{m+r-1}+2^j)+(D-2^j)=D+2^{m+r-1}$. Hence $L_r=(D+2^{m+r-1})L_{r-1}$ and $L_0=1$. Dividing by $Z=D\prod_{i=0}^{t-2}(D+2^i)$ gives $c_r/(D\prod_{i=0}^{t-r-2}(D+2^i))=F_{t-r}$. □

Theorem 10. *For all $a,b\ge 0$ with $t=a+b\ge 1$ and all $n\ge 1$,*

$$\lambda_{\max}\big(X^{(a,b)}\big)=F_{t-w},\qquad w=\min(a,b)+\kappa(|a-b|).$$

Proof. By Lemma 5 it suffices to show $\rho(M)=F_{t-w}$. Levels are dimensions of totally isotropic subspaces, so every level is at most w (Proposition 6); level w is attained by U_N of Lemma 8 for a maximal isotropic N, since $N_L(U_N)=N$. Since $F_{s+1}/F_s=(2^{s-1}+1)/(D+2^{s-1})\le 1$, F_{t-r} is nondecreasing in r, so by Lemmas 7–9 every column sum of M is at most F_{t-w}, and $\rho(M)\le F_{t-w}$. If $(0,z)\in U'$ then $(0,z)\in U\circ U'$, because $(0,0)\in U$; so $N_R(U\circ U')\supseteq N_R(U')$ and the level never decreases under left multiplication (recall $\dim N_L=\dim N_R$ for every element of Σ, Section 3). Ordering levels increasingly, M is block lower-triangular, and the columns at level w send all of their mass into level w. The diagonal block M_{ww} therefore has every column sum equal to F_{t-w}, so $\mathbf{1}^{\mathsf T}M_{ww}=F_{t-w}\mathbf{1}^{\mathsf T}$ and F_{t-w} is an eigenvalue of M_{ww}, hence of M. Thus $\rho(M)=F_{t-w}$. □

Corollary 11. *For $1\le T\le 3$, $k\ge 1$ and $n\ge 1$,*

$$P_{\mathrm{stab}}(n;T,k)=\binom{D+T-1}{T}F_{T+k-\kappa(k)}.$$

Proof. Lemma 1 and Theorem 10 with $(a,b)=(T+k,T)$, $t=2T+k$ and $w=T+\kappa(k)$. □

6 Consequences

When the universal cloner is optimal. Let $m=T+k$. Since $\kappa(k)\ge 0$ and F_s is decreasing in s, Corollary 11 gives $P_{\mathrm{stab}}/D_T=F_{m-\kappa(k)}\ge F_m$. The frame-potential inequality gives $F_m\ge 1/D_m$, with equality exactly when the ensemble is a projective m-design; stabilizer states are 3-designs but not 4-designs [5, Cor. 1], so $F_m>1/D_m$ for every $m\ge 4$. Conversely, when $m\le 3$ we have $\kappa(k)=0$ and $F_m=1/D_m$. Therefore the universal cloner is optimal exactly when $T+k\le 3$, in agreement with the design criterion of [7]; the first strict gap appears at $T+k=4$ (e.g. $1/8$ against $4/35$ at $n=2$, $1\to 4$).

Plateaus. The optimum is unchanged from k to $k+1$ whenever $\kappa(k+1) = \kappa(k) + 1$, i.e. for $k+1 \equiv 4, 6, 7, 0 \pmod 8$: for instance $1 \to 4$ and $1 \to 5$ have the same value, as do $1 \to 6$ through $1 \to 9$. Producing one more copy is free exactly when a longer doubly-even code becomes available.

Many copies. For $s \geq n+1$, cancelling powers of two gives the exact identity

$$F_s = \frac{1}{N_n} \prod_{i=s-1-n}^{s-2} \left(1 + 2^{-i}\right),$$

so $F_s \to 1/N_n$ and $P_{\text{stab}}(n; T, k) \to D_T/N_n$ as $k \to \infty$. For $T \leq 3$ the T-copy states form a tight frame for Sym^T (3-design property), so D_T/N_n is the optimal probability of identifying the state from T copies, attained by the pretty-good measurement (any POVM $\{E_S\}$ succeeds with probability $\frac{1}{N_n}\sum_S \langle S^{\otimes T}|E_S|S^{\otimes T}\rangle \leq \frac{1}{N_n}\operatorname{tr}\sum_S E_S = D_T/N_n$). Cloning many copies thus becomes identification.

Scope. The results concern average global fidelity; by Lemma 1 they are also the worst-case optima over stabilizer states. Theorem 10 holds for every signature, but the reduction to cloning uses $T \leq 3$; for $T \geq 4$ the Clifford group is not a unitary 4-design [6] and the optimal dual need not be scalar.

Part II. The price of measuring first

7 Measure-and-prepare cloning

A *measure-and-prepare* cloner measures the T input copies with a POVM $\{E_x\}$ and, on outcome x, prepares a state σ_x of $m = T+k$ systems; these are exactly the entanglement-breaking channels [18]. Let $P_{\text{EB}}(n; T, k)$ be the largest average global fidelity they achieve. Throughout this part $z = |0^n\rangle$,

$$K_{T,m} = \mathbb{E}_S\, |S\rangle\langle S|^{\otimes T} \otimes |S\rangle\langle S|^{\otimes m} \quad \text{on } \text{Sym}^T(\mathbb{C}^D) \otimes \text{Sym}^m(\mathbb{C}^D),$$
$$B_{T,m} = \mathbb{E}_S\, |\langle z|S\rangle|^{2T}\, |S^{\otimes m}\rangle\langle S^{\otimes m}|.$$

Here $\mathbb{E}_S$ is the uniform average over $\text{Stab}(n)$, and $B_{T,m} = (\langle z^{\otimes T}| \otimes I) K_{T,m} (|z^{\otimes T}\rangle \otimes I)$. Note that $K_{T,m}$ carries no complex conjugation, unlike the operator $X^{(T+k,T)}$ of Part I; the two must not be interchanged without conjugating the input factor.

Lemma 12 (Reduction). *For every $T \geq 1$, $P_{\text{EB}} \leq D_T\, \Lambda$ with $\Lambda = \max\{\langle \phi \otimes \chi | K_{T,m} | \phi \otimes \chi\rangle :\ \phi \in \text{Sym}^T,\ \chi \in \text{Sym}^m$ unit$\}$. For $1 \leq T \leq 3$ equality holds.*

Proof. The fidelity of $\{E_x, \sigma_x\}$ is $\sum_x \operatorname{tr}[(E_x \otimes \sigma_x) K_{T,m}]$. Compressing the effects to Sym^T and refining them into rank-one effects $c_x |\phi_x\rangle\langle\phi_x|$ with pure outputs χ_x (by convexity) gives at most $\sum_x c_x \Lambda = D_T \Lambda$, since $\sum_x c_x = \dim \text{Sym}^T = D_T$. For $T \leq 3$ let (ϕ_*, χ_*) attain Λ. Because the Clifford group is a unitary 3-design [13, 14], the effects $(D_T/|\mathcal{C}|)\, C^{\otimes T} |\phi_*\rangle\langle\phi_*| C^{\dagger \otimes T}$, $C \in \mathcal{C}$, sum to the identity on Sym^T; preparing $C^{\otimes m}|\chi_*\rangle$ on outcome C achieves $D_T \Lambda$ by invariance of $K_{T,m}$. $\square$

Two benchmark strategies. Measuring with the pretty-good measurement, whose effects $E_S = (D_T/N_n)|S^{\otimes T}\rangle\langle S^{\otimes T}|$ form a POVM on Sym^T for $T \leq 3$, and preparing m copies of the guess gives exactly

$$P_{\text{PGM}}(n; T, k) = D_T\, F_{2T+k}, \tag{1}$$

since the success weight is $(D_T/N_n)|\langle S|S'\rangle|^{2T}$ and the output overlap is $|\langle S|S'\rangle|^{2m}$. Keeping the same measurement but preparing, on each outcome, the best m-copy state (the *pretty-good-measurement strategy* below) gives the *seed* value $P_{\text{seed}} = D_T \lambda_{\max}(B_{T,m})$; this is the restriction of Λ to stabilizer seeds $\phi = S^{\otimes T}$. Comparing (1) with Corollary 11,

$$\frac{P_{\text{stab}}}{P_{\text{PGM}}} = \frac{F_{T+k-\kappa(k)}}{F_{2T+k}} = \prod_{i=T+k-\kappa(k)-1}^{2T+k-2} \frac{D+2^i}{2^i+1} > 1.$$

The PPT relaxation. Write Γ for the transpose of the Sym^T factor and $P_{\text{PPT}} = \max\{\mathrm{tr}(K_{T,m}\rho) : \rho \succeq 0,\ \mathrm{tr}\,\rho = 1,\ \rho^\Gamma \succeq 0\}$ over states on $\mathrm{Sym}^T \otimes \mathrm{Sym}^m$. Product states are feasible, so $\Lambda \le P_{\text{PPT}}$, and for any $W \succeq 0$ with $S := yI - K_{T,m} - W^\Gamma \succeq 0$ we have $\mathrm{tr}(K\rho) = y - \mathrm{tr}(S\rho) - \mathrm{tr}(W\rho^\Gamma) \le y$ (weak duality). Upper bounds below are certificates (y, W, S) of this form.

8 The identification regime

The first question is when re-preparing helps at all. Stabilizer amplitudes have equal moduli on their support and relative phases in $\{\pm 1, \pm i\}$ [19]; fourth powers erase these phases.

Theorem 13 (Onset at four copies)**.** *For every $n \ge 1$ and $T \ge 1$: if $m \le 3$ then $\lambda_{\max}(B_{T,m}) = F_{T+m}$, attained only by $z^{\otimes m}$; if $m \ge 4$ then $\lambda_{\max}(B_{T,m}) > F_{T+m}$. Hence for $1 \le T \le 3$ the seed strategy strictly beats P_{PGM} exactly when $T + k \ge 4$.*

Proof. Let $v = z^{\otimes m}$; then $\langle v|B|v\rangle = \mathbb{E}_S|\langle z|S\rangle|^{2(T+m)} = F_{T+m}$. *($m \ge 4$.)* Let $x = |10\cdots 0\rangle$, $u = x^{\otimes 4} \otimes z^{\otimes(m-4)}$ and $\alpha_y = \langle y|S\rangle$. Then $u \perp v$ and $\langle u|B|v\rangle = \mathbb{E}_S|\alpha_0|^{2(T+m-4)}(\alpha_x\bar{\alpha}_0)^4 \ge 0$, each term being real and nonnegative by the phase structure; the term of $|+\rangle \otimes z_{n-1}$ is positive. So v is not an eigenvector and $\lambda_{\max}(B) > \langle v|B|v\rangle$. *($m \le 3$.)* B commutes with $U^{\otimes m}$ for every diagonal Clifford U fixing z; applying the phase gate on one qubit shows $\langle w|B|v\rangle = 0$ unless, on every qubit, the number of copies carrying a 1 is divisible by 4, which for $m \le 3$ forces $w = v$; so $Bv = F_{T+m}v$. By the moment formula of [8, Thm. 5.3], $B = Z_{n,t}^{-1} \sum_{L \in \Sigma_{t,t}} R_z(L)$ with $t = T + m$ and $R_z(L)$ the compression of $R(L)$ to z on the input slots. If a column of $R_z(L)$ had two nonzero entries x', x'', then $(0^T, x' + x'', 0) \in L$, and total isotropy forces $|x' + x''| \equiv 0 \pmod 4$, impossible for $m \le 3$ unless $x' = x''$; similarly for rows. Each $R_z(L)$ is therefore a partial permutation, hence a contraction, and $\|B\| \le |\Sigma_{t,t}|/Z_{n,t} = F_t$; with $Bv = F_t v$ this gives $\lambda_{\max}(B) = F_t$. For uniqueness, let $Bu = F_t u$ with $\|u\| = 1$. Since B is F_t times an average of the contractions $R_z(L)$, equality forces $R_z(L)u = u$ for every L. For the relation L_j given by the transposition of the input slot with output slot j, compressing the input onto z makes $R_z(L_j)$ the projector onto states whose j-th output copy is z. Hence u is supported on basis vectors with every output copy equal to z, so $u \propto z^{\otimes m}$. □

Theorem 14 (Seed values at four output copies)**.** *Let $m = 4$, $T \in \{1, 2, 3\}$, $e = z^{\otimes 4}$ and $f = (D-1)^{-1/2} \sum_{x \ne 0} |x\rangle^{\otimes 4}$. For every $n \ge 1$, $\lambda_{\max}(B_{T,4})$ is simple, its eigenvector lies in* $\mathrm{span}\{e, f\}$*, and it equals the larger eigenvalue of*

$$M_T = \begin{pmatrix} F_{T+4} & \frac{F_{T+3} - F_{T+4}}{\sqrt{D-1}} \\ \frac{F_{T+3} - F_{T+4}}{\sqrt{D-1}} & \frac{F_{T+2} - 2F_{T+3} + F_{T+4}}{D-1} \end{pmatrix}.$$

Explicitly, with $Z_5 = D(D+1)(D+2)(D+4)(D+8)$, $Z_6 = Z_5(D+16)$, $Z_7 = Z_6(D+32)$,

$$\lambda_1 = \frac{3\big(D+48+\sqrt{D^2+16D+1664}\big)}{Z_5}, \qquad \lambda_2 = \frac{15\big(D+160+\sqrt{D^2+32D+20992}\big)}{Z_6},$$
$$\lambda_3 = \frac{135\big(D+576+\sqrt{D^2+64D+296960}\big)}{Z_7}.$$

Proof outline. M_T is B compressed to $\{e,f\}$: by the phase structure, $\sum_{x\neq 0}\alpha_x^4 = (2^r-1)\alpha_0^4$ when $\alpha_0 \neq 0$ and the support of S has dimension r, and $\mathbb{E}_S|\alpha_0|^{2u} = F_u$. This gives the lower bound. For the upper bound, B splits into blocks indexed by characters of the diagonal Clifford group fixing z; inside a block all entries are nonnegative, so each block has a Perron eigenvector. The trivial block's invariant subspace is $\mathrm{span}\{e,f\}$ ($\mathrm{GL}(n,2)$ acts transitively on $x \neq 0$, and the only doubly-even word of length 4 is 1111). A finite classification of the 902 totally isotropic subspaces involved reduces every other block, for every n, to two class types and singletons, whose Perron roots are strictly below $\lambda_{\max}(M_T)$ by explicit polynomial inequalities in D and 2^T, plus separate boundary cases. The classification and the polynomial inequalities are computer-assisted and were checked independently (Appendix A). □

Theorem 15 ($1 \to 4$ for every n)**.** *For every* $n \geq 1$,

$$P_{\mathrm{EB}}(n;1,3) = D\lambda_1 = \frac{3\big(D+48+\sqrt{D^2+16D+1664}\big)}{(D+1)(D+2)(D+4)(D+8)},$$

attained by the pretty-good measurement followed by the optimal re-preparation.

Proof. Lower bound: Theorem 14 (the compression alone suffices). Upper bound: a dual certificate for the PPT relaxation. The relevant operators live in the span of the six orbit sums $b_0,\dots,b_5$ of the relation operators $R(L)$, $L \in \Sigma_{5,5}$, under permutations of the four output coordinates on each side, and their images $b'_k = \Gamma(b_k)$. Label them by orbit size: b_4 (24 elements) is the orbit of the identity relation, b_5 (96) the orbit containing the graph of the transposition exchanging the input coordinate with an output coordinate, b_1 (6), b_2 (the other orbit of size 96), and b_0, b_3 the two remaining orbits of size 24, which are exchanged by transposition. With $s = \sqrt{D^2+16D+1664}$, $c = (D+40+s)/64$ and $Z = Z_5$, put

$$Z\,W = -\tfrac{1}{2}(b'_0+b'_1+b'_3) - c\,b'_2 + 4c\,b'_4 + (c-1)\,b'_5, \qquad S = \lambda_1 P - K_{1,4} - \Gamma(W),$$

where $P = b_4/24$ is the unit and $K_{1,4} = Z^{-1}\sum_k b_k$. Both algebras are closed under multiplication and transposition, and for $D \geq 4$ the b_k are linearly independent. The characteristic polynomials of left multiplication by ZW and by ZS both equal $t^3(t-\alpha)(t-\tau)^2$ with $\alpha = \frac{15}{8}(D+40+s)$ and $\tau = \frac{3}{8}\big(D^2-24D+64+(D+8)s\big) > \frac{3}{4}(D^2-4D+64) > 0$; hence $W, S \succeq 0$ for $D \geq 4$. At $D = 2$ the same operators are checked by exact principal minors. The dual value is λ_1, so $\Lambda \leq \lambda_1$, and Lemma 12 gives the claim. □

Corollary 16. *For* $1 \to 4$ *and every* n, $P_{\mathrm{stab}}/P_{\mathrm{EB}} = 10(D+8)/\big(D+48+\sqrt{D^2+16D+1664}\big)$, *which increases from* 1.096 *at* $n=1$ *to the limit* 5 *and is always below* 5.

Proof. $P_{\mathrm{stab}} = D\,F_4 = 30/((D+1)(D+2)(D+4))$ by Corollary 11; divide by Theorem 15. With $x = D+8$ and $s = \sqrt{x^2+1600} = \sqrt{D^2+16D+1664}$ the ratio is $R = 10x/(x+40+s)$, and $dR/dx = 10\,(40+1600/s)/(x+40+s)^2 > 0$. Since $s > x$, $R < 10x/(2x) = 5$, and $R \to 5$ as $x \to \infty$. □

9 The Bell-sampling regime

With two or more input copies a different measurement is available. *Bell sampling* measures qubit i of one copy together with qubit i of another in the Bell basis; outcomes are labelled by Pauli operators P.

Proposition 17 (Bell sampling)**.** *For a stabilizer state only symmetric Paulis ($P^{\mathsf{T}} = P$) occur, each with probability $1/D$ on a coset of D outcomes [11, Lemma 2 and the analysis following it]. On outcome P prepare the normalized symmetrization $\chi_P \propto \Pi^{(4)}_{\mathrm{sym}}(\Phi'_P \otimes \Phi'_P)$, where $\Phi'_P = \sum_x P|x\rangle|x\rangle$. This output is optimal for each outcome, and for every n the strategy achieves the $2 \to 4$ fidelity*

$$P_{\mathrm{Bell}} = \frac{3}{D(D+2)},$$

and the same value for $3 \to 4$ when one input copy is discarded.

Proof outline. For each occurring outcome the conditional output operator has largest eigenvalue $g(D) = 6/(D^2(D+1)(D+2))$: the Rayleigh quotient of χ_P (with $\|\Pi^{(4)}_{\mathrm{sym}}(\Phi'_P \otimes \Phi'_P)\|^2 = D(D+2)/3$) gives g from below, and exact row sums of a nonnegative Gram matrix give g from above, via a q-binomial count of real stabilizer states. Summing over the $D(D+1)/2$ symmetric outcomes gives $3/(D(D+2))$. □

Theorem 18 (Two regimes for $2 \to 4$)**.**

$$P_{\mathrm{EB}}(n;2,2) = \begin{cases} D_2\lambda_2 & n \le 3, \\ \dfrac{3}{D(D+2)} & n \ge 4, \end{cases}$$

with values $(9+\sqrt{65})/32$, $(41+\sqrt{1321})/384$ and $(7+\sqrt{37})/256$ at $n = 1, 2, 3$. The two strategies cross at $D_* = 13.0695596789\ldots$, the unique root in $[2,\infty)$ of $D^4 + 36D^3 - 232D^2 - 4096D - 16384$. For $n \ge 4$, $P_{\mathrm{stab}}/P_{\mathrm{EB}} = 5D/(D+4) < 5$.

Proof. For $n \le 3$ the upper bound is an exact PPT dual certificate at each n and the lower bound is Theorem 14. For $n \ge 4$ the lower bound is Proposition 17; the upper bound is a PPT dual certificate valid for every real $D \ge 16$, built in the 21-dimensional algebras of orbit sums of $\Sigma_{6,6}$ (4590 relations) under $(S_2 \times S_4) \times (S_2 \times S_4)$, with dual value $g(D)$ and positivity proved symbolically in D (Appendix A). Multiplying by D_2 gives $3/(D(D+2))$. The crossover follows from comparing the closed forms. □

For four or more qubits Bell sampling is therefore optimal and the pretty-good-measurement strategy is suboptimal: the best measure-and-prepare cloner learns stabilizer information from pairs of copies. With a single input copy Bell sampling is unavailable, consistent with the single regime found for $1 \to 4$ (Theorem 15). In both cases the advantage of coherent cloning stays below a factor of 5.

10 Stabilizer and non-stabilizer outputs

Proposition 19. *Let $T \ge 1$, $m \ge 4$, $s = T + m$ and $\chi = D^{-1/2}\sum_x |x\rangle^{\otimes 4} \otimes z^{\otimes(m-4)}$, a stabilizer state. Then $\langle\chi|B_{T,m}|\chi\rangle = F_{s-2}/D$, which exceeds F_s exactly when $(D-1)(D-2^{2s-5}) > 0$; with j disjoint blocks of this kind ($m \ge 4j$) one obtains F_{s-2j}/D^j. For $m > 4$, χ is a valid output state on the m systems but not permutation-symmetric; Rayleigh quotients of any output state bound $\lambda_{\max}(B_{T,m})$ from below.*

Proof. For $\alpha_0 \neq 0$, $\sum_x \alpha_x^4 = 2^r \alpha_0^4$, so $|\langle\chi|S^{\otimes m}\rangle|^2 = D^{-1}2^{2r}|\alpha_0|^{2m}$ and $\mathbb{E}_S|\alpha_0|^{2T}D^{-1}2^{2r}|\alpha_0|^{2m} = F_{s-2}/D$. The comparison reduces to $D^2 - D(2^{2s-5}+1) + 2^{2s-5} > 0$. □

So, for $1 \leq T \leq 3$, re-preparing with an ordinary stabilizer output already beats copying the guess once $n > 2s-5$ (for example $n \geq 6$ at $1 \to 4$). At $n = 1$ the situation differs: for $m = 4, 5$ an exhaustive exact search over all stabilizer outputs (36,720 and 2,423,520 states; computer-assisted, with the $m = 5$ search not independently re-verified) finds none better than P_{PGM}, so the gain of Theorem 13 there requires non-stabilizer outputs.

11 Three input copies

For $3 \to 4$ the picture is incomplete. Proved: $P_{\mathrm{EB}} = D_3\lambda_3$ at $n = 1, 2$ (exact certificates); Bell sampling on two of the three copies beats the seed strategy exactly when $D > 64.2342\ldots$ (the root in $[2, \infty)$ of $D^6 + 74D^5 - 5992D^4 - 155200D^3 - 1830976D^2 - 7143424D - 8388608$), so the pretty-good-measurement strategy is not optimal for $n \geq 7$; and for $n = 4, 5, 6$ the three-copy seed value exceeds $3/(D(D+2)) = P_{\mathrm{EB}}(n; 2, 2)$, so the third copy strictly helps. These three comparisons use only closed forms proved above and were checked by a second system (the crossover polynomial, its unique root in $[2, \infty)$, and the signs at $D = 2^n$). Open: the exact value of $P_{\mathrm{EB}}(n; 3, 1)$ for $n \geq 3$. Computer-assisted PPT bounds for $4 \leq n \leq 10$ (exact rational dual certificates produced by one system, not independently re-verified) exist, and at $n = 6$ an exact rational PPT state from the same computation (likewise not independently re-verified) exceeds the Bell value, so no PPT dual with the Bell value exists there; whether the PPT relaxation is tight for $3 \to 4$ remains open.

Part III. Open problems

12 Discussion and open problems

Coherent cloning of stabilizer states interpolates between Werner's universal value (few copies) and optimal identification (many copies), with the transition controlled by doubly-even codes. Measuring first costs a bounded factor in the cases solved here: below 5 for $1 \to 4$ and $2 \to 4$ at every n, with the optimal measurement for $2 \to 4$ switching from the pretty-good measurement to Bell sampling as n grows. The same $\mathbb{Z}_4$ arithmetic, through the phases $\pm 1, \pm i$ and the weight-4 doubly-even word, controls both the coherent optimum and the onset of the measure-and-prepare gain. Open problems:

1. the exact measure-and-prepare optimum for $3 \to 4$ at $n \geq 3$, and for five or more output copies (at $m = 6, 7$ new classes from the doubly-even codes of lengths 6 and 7 appear);
2. coherent cloning for $T \geq 4$, where the Clifford group is not a unitary 4-design [6];
3. an explicit optimal coherent channel, and other figures of merit such as the trace distance used in [2];
4. qudit stabilizer states, where Bell sampling generalizes [20] and the relevant quadratic forms change with the prime;
5. whether copying limits of other state families are governed in the same way by an arithmetic invariant of their relation algebras.

Related results concern exact stabilizer identification [12], $O(n)$-copy learning by Bell sampling [11] and Bell difference sampling [8, 21], sample-optimal learning bounds [22], cloning sample complexity [3, 4, 23], exact single-copy fidelity regions for asymmetric $1 \to 2$ cloning of orthogonal-group orbits [24], cloning of a pair of known states with a fixed failure rate [25], the quantumness of state sets under measure-and-prepare processing [26], fixed-target forgery thresholds for stabilizer public keys [27], and the structure of the Clifford commutant [6, 8, 28, 29].

Verification. Part I is proved analytically; its computer checks are supplementary. Exact symbolic certificates (the spectrum of the orbit quotient of M, valid for every n) were computed for the 15 signatures (a, b), $a \geq b$, with $3 \leq t \leq 7$ other than the trivial $(3, 3)$; Lemmas 7 and 8 were verified exactly over all 9,845,550 relations for each of the five signatures with $t = 8$; a separate implementation confirmed Lemmas 7 and 8 and $\rho(M) = F_{t-w}$ for $t \leq 5$ at $n = 1, 2, 3$; and an exhaustive search confirmed Proposition 6 for all 54 signatures with $t \leq 9$. Part II combines analytic arguments with exact certificates; Appendix A lists each certificate and how it was checked independently.

A Certificates and independent checks

The certificates supporting the theorems of Part II are exact (rational or quadratic-field arithmetic) unless marked numeric, and are included in the file `paper3_v2_certificates.zip` archived with this version, whose SHA-256 is

`236f7e4c5405e8e4030d476958e51aae5fb7b2fd2349c3ec9ddcb882cddf054f`

Its `README.md` gives, for each item, the folder, the command, what it checks, and who produced and who independently checked it. `run_all.sh` re-runs every bundled check at sizes that fit in about 1 GB of memory per process; `run_full_checks.ps1` runs the three larger checks (the $n = 3$ certificate for $2 \to 4$ and the full Bell-sampling checks at $n = 4, 5$). Both scripts judge a check as passed only if it exits normally and its log contains the expected success marker, and both exit with a nonzero status if any check fails. `logs_expected/` holds the logs of these runs, and `MANIFEST_SHA256.txt` lists the hash of every file. Not bundled: the exhaustive stabilizer-output search of Section 10 (reported; for $m = 5$ not independently re-verified).

Statement	Certificate	Independent check
Thm. 13 ($m \leq 3$ and $m \geq 4$)	analytic	second system, line by line
Thm. 14, dominance step	`thm14_seed_values/`: classification of the 902 isotropic subspaces; polynomial inequalities in D, 2^T	re-enumerated (same counts, no violations); inequalities recomputed
Thm. 15, $D \geq 4$	`thm15_1to4_all_n/`: six-dimensional algebras, explicit W, characteristic polynomials	algebra rebuilt from the definitions; polynomials re-derived symbolically
Thm. 15, $D = 2$	`thm15_1to4_all_n/grok_D2_check/`: exact principal minors	recomputed in $\mathbb{Q}(\sqrt{17})$
Thm. 18, $n = 1, 2, 3$	`thm18_2to4/n1,n2,n3/`: PPT dual certificates (dimensions up to 11,880)	separate checkers written from the README (Claude: $n = 1, 2$; ChatGPT: $n = 3$)
Thm. 18, $n \geq 4$	`thm18_2to4/all_D/`: dual in the 21-dimensional algebras of $\Sigma_{6,6}$, positivity over $\mathbb{Q}(D)$	all 4590 relations re-validated; every structure constant regenerated (no mismatch); certificate re-run
Prop. 17	`prop17_bell/`: analytic; exact counts for $n \leq 5$	second system, line by line
Section 11 comparisons	closed forms; crossover polynomial and signs	root and signs confirmed by a second system
Prop. 19, $n = 1$ exhaustive search	exact enumeration ($m = 4, 5$); not bundled	$m = 4$ reproduced numerically; $m = 5$ not re-verified

Use of AI tools

This work used several AI systems; the author directed it and takes full responsibility for its content. *Part I.* **Grok Bot** (SpaceXAI) proposed the closed form, proved Theorem 10 with Lemmas 7–9 and the upper bound of Proposition 6, verified the reduction of Lemma 1, and produced the exact certificates and the $t = 8$ check. **Claude** (Anthropic; Claude Opus 5.5) derived the reduction of Lemma 1, wrote the proof of Lemma 3, audited every proof, wrote the separate implementation and the independent numerical checks, reproduced the certificates, and drafted the manuscript. **ChatGPT** (OpenAI) refereed the proofs, finding a false operator identity and an overstated domain in earlier versions, and supplied the short proof of the spectral inclusion and the attainment argument in Lemma 5, the code constructions in Proposition 6, and the choice of w in Lemma 8. *Part II. Proposed:* Claude proposed the value (1), the two-dimensional compression of Theorem 14 and the Bell-sampling strategy for $2 \to 4$. *Proved:* Grok Bot proved Lemma 12, Theorem 14, the $m \leq 3$ half of Theorem 13 and Propositions 17 and 19, and constructed the dual certificates of Theorems 15 and 18 (including the $n = 1, 2$ certificates) and derived the crossover polynomial of Theorem 18; ChatGPT proved the $m \geq 4$ half of Theorem 13 for $T \leq 3$, Grok Bot extended it to every $T \geq 1$ and Claude checked it; Claude proved the matching lower bound of Theorem 15. *Independently checked (each with the checker's own code):* Claude rebuilt the relation algebras of Theorems 15 and 18 (every structure constant), re-derived the characteristic polynomials of Theorem 15, and checked the $n = 1, 2$ certificates; ChatGPT checked the $D = 2$ minors of Theorem 15, the $n = 3$ certificate of Theorem 18 and the classification in Theorem 14, confirmed the Section 11 crossover and comparisons, and refereed every Part II result. Grok Bot derived the Section 11 comparisons and produced its computer-assisted PPT bounds. *Reported, not independently reproduced:* the $3 \to 4$ PPT bounds of Section 11 and the $m = 5$ exhaustive search of Section 10. **Perplexity** and **Consensus** carried

out the prior-art searches. A further system's reported verification was discarded after its code was found not to compute what it reported.

Data and code availability

Supporting code, certificates, independent checkers, logs and reproduction instructions are provided in two archives, `paper3_code.zip` (Part I) and `paper3_v2_certificates.zip` (Part II), with SHA-256

```
paper3_code.zip  de2fc42b8bc4fa710b8506de2001d475a8d8b4a011ed0df72c693110a5d417e0
paper3_v2_certificates.zip  see Appendix A
```

On release, they are deposited with this preprint's Zenodo record (version 2) and in the folder `paper3/` of `https://github.com/Weldog/sidon-phase-cloning` (release tag `paper3-v2`); any copy can be checked against these hashes.

Competing interests

The author declares no competing interests. This work received no external funding.

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